RESEARCH PROJECT • ASSIGNMENT 2

Probabilistic Language Modeling and Input Noise

Extend your verified pipeline into the first classical-baseline experiment for your cumulative ACL-style paper.

TEAM	PHASE	MODELS	SUBMIT
Same 2 students	2 — Baselines	Unigram, bigram, mixture	Paper v0.2 + repo

1. Purpose and research question

Assignment 1 established your WikiText subcorpus, character-noise functions, Minimum Edit Distance (MED) tests, and fixed WordPiece tokenizer. Reuse this foundation to measure how clean-trained n-gram models score clean and noisy text. These preliminary validation results become the classical baseline in your semester-long ACL-style paper; neural robustness evidence comes later.

RESEARCH QUESTION How do character-noise type and intensity change token coverage, fragmentation, and predictive behavior under fixed whitespace and WordPiece tokenization?

2. What carries forward

Retain from Assignment 1	Add in Assignment 2
Corpus audit, IDs, splits and source hashes	A frozen evaluation sample and boundary policy
Insertion, deletion, substitution and manifests	A reproducible grid of noise rates and seeds
MED tests and 20 paired diagnostic examples	Probability tests and aggregate results
Fixed WordPiece settings and ethics statement	Whitespace comparison and cumulative paper draft

Learning outcomes: estimate conditional probabilities, handle unseen events, compute loss and perplexity in consistent units, interpret paired experiments, and write claims supported by saved evidence.

SCOPE: Reuse the BERT-compatible tokenizer from Assignment 1. The predictive models here are n-grams; BERT representation analysis and neural evaluation come later.

RESEARCH PROJECT • ASSIGNMENT 2

Your experiment at a glance

One team of two students builds a reproducible baseline for the same semester-long paper.

What you will do

Reuse your Assignment 1 subcorpus and noise code. Learn token counts from clean training text, then score a fixed validation sample in clean and corrupted forms. Apply noise to text before tokenization. Reuse exactly the same noisy records for whitespace and the fixed pretrained WordPiece tokenizer.

A unigram uses no preceding token; a bigram uses one. The third model combines unigram, bigram and trigram predictions, with the trigram component using two preceding tokens. Each model assigns probabilities to the actual next tokens. You are measuring prediction quality, not asking the models to generate sentences.

How many scores you will produce

Quantity	Calculation	Total
Model and tokenizer configurations	3 models × 2 tokenizers	6
Noisy validation datasets	3 operations × 4 rates × 3 seeds	36
Individual clean perplexity scores	6 configurations × 1 clean sample	6
Individual noisy perplexity scores	6 configurations × 36 datasets	216
All individual perplexity scores	6 clean + 216 noisy	222
Noisy summary rows	6 configurations × 3 operations × 4 rates	72

Operations: insertion, deletion and substitution. Nonzero rates: 1%, 5%, 10% and 20%. Corruption seeds: 11, 23 and 47. The clean baseline is shared across seeds and operations; do not count it repeatedly.

How to report and interpret the results

Report 6 clean perplexities and 72 noisy summary rows, each containing the mean, minimum and maximum of its three seed-specific perplexities. First compute each corpus perplexity from total negative log probability and total scored targets; do not average sentence perplexities. Retain the loss values and diagnostics required in Section 9.

Compare clean and noisy scores within each tokenizer. Raw perplexity cannot rank different tokenizers fairly. A lower noisy score can result from unknown-token collapse or changed targets. Explain the observed pattern rather than assuming noise must always increase perplexity.

These results supply the classical Methods, Experimental Setup and preliminary Results for paper version 0.2. Later neural experiments will reuse your data and noise procedures, while adding evidence beyond this baseline.

RESEARCH PROJECT • ASSIGNMENT 2

Corpus and character-noise protocol

Student A extends the existing pipeline; both students approve the saved configuration.

3. Data continuity and split discipline

Reuse the Assignment 1 WikiText subcorpus, slice rules and supplied splits; the default is wikitext-2-raw-v1. Follow Section 21 of this assignment for bounded sampling. Document any approved alternative, dataset migration or sample change and its reason; regenerate affected diagnostics. Never silently replace data.

Count n-grams on clean training data. Select interpolation weights on clean validation data, then freeze settings before the validation noise grid. All Assignment 2 findings are preliminary validation evidence. Do not evaluate test conditions now. Freeze the semester analysis plan before the final classical and neural test runs. If test outcomes were already inspected, notify the instructor and document the exposure; do not call them untouched.

4. Required corruption conditions

Operation	Default rule on original eligible characters
Deletion	Independently delete an eligible character with probability p.
Substitution	With probability p, replace a character with a different lowercase a–z character, sampled uniformly.
Insertion	With probability p, insert one uniformly sampled lowercase a–z character immediately after an eligible character.

Eligible characters are ASCII letters a–z and A–Z. Preserve spaces, punctuation, digits, record boundaries, and corpus sentinel strings. Sample once from original positions; do not corrupt newly inserted characters. Retain different Assignment 1 rules only if explicitly recorded and applied consistently across all comparisons.

Core rates: p = 0, 0.01, 0.05, 0.10, 0.20. Use seeds 11, 23, and 47 for each nonzero condition. *Optional Extensions*: Rates 0.30 and 0.50, keyboard substitutions, and adjacent swaps. A swap is not a unit-cost Levenshtein operation.

5. Pairing and provenance

Generate each noisy text once and pass that exact text to both tokenizers. Save source-document and record IDs, split, seed, operation, requested rate, edit log, eligible count, MED, normalized MED, and source/configuration hashes. Reproduce the 20 Assignment 1 diagnostic pairs; they are not the aggregate evaluation sample.

RESEARCH PROJECT • ASSIGNMENT 2

Models and probability definitions

Student B extends the tokenizer baseline into a tested counting and scoring engine.

6. Tokenizers and vocabulary

Keep the exact pretrained WordPiece tokenizer and revision from Assignment 1. If the checkpoint was not recorded, resolve it before running; the course default is google-bert/bert-base-uncased. Record casing, accents, punctuation handling and versions. Compare with Python text.split(), retaining case and punctuation. This compares complete pipelines, not segmentation alone.

Whitespace output support V contains training tokens occurring at least twice, <UNK>, and an end-of-record symbol. Map other tokens to <UNK>. WordPiece support keeps all vocabulary entries except [PAD], [CLS], [SEP], [MASK] and [unused...] entries, and adds an end symbol; retain [UNK] and all other unseen entries. Freeze V and report |V| for each tokenizer.

Use every retained nonempty clean record as one sequence. Reset context; use one start symbol for bigrams and two for trigrams, and score one end symbol. Start symbols are context only. If noise deletes all text, retain the pair and score its end symbol. Note: N-grams use no BERT wrappers, padding or 512-token truncation.

7. Required model configurations

$$P_MLE(w \mid h) = \frac{C(h,w)}{C(h)} \quad when\ C(h) > 0$$

$$q_k(w \mid h) = \frac{C(h,w) + 1}{C(h) + |V|}$$

Here k is n-gram order and h contains the last k−1 tokens. C(h,w) counts training events in which w follows h; C(h) sums these counts over V. For q_1 the history is empty. All orders share one support per tokenizer. Unseen histories become uniform under add-one smoothing. See §20 for all notation.

$$P_mix(w \mid h) = \lambda_1 q_1(w) + \lambda_2 q_2(w \mid last\ token) + \lambda_3 q_3(w \mid last\ two\ tokens)$$

Score three configurations per tokenizer: add-one unigram q_1, add-one bigram q_2, and the interpolated unigram/bigram/trigram model P_mix. The third configuration is not standalone q_3. Score observed next tokens from left to right; generating sentences is not required. Demonstrate unsmoothed MLE separately on a small test case; retain any infinite loss from zero-probability events.

Choose nonnegative λ_1, λ_2, λ_3 summing to 1 on a grid with increments of 0.25, minimizing clean-validation loss separately for each tokenizer. Break ties deterministically, record the rule, and freeze the weights for all noise conditions. Use sparse counts and compute probabilities on demand.

RESEARCH PROJECT • ASSIGNMENT 2

Measurement and interpretation

Keep probability units and the meaning of each comparison explicit.

8. Loss and perplexity

$$H_bits = -\frac{1}{T}\sum_i log_2 P(w_i \mid h_i)$$

$$PP = 2^{H_bits}; \quad H_bits = \frac{H_nats}{ln\ 2}$$

T is the total number of scored targets, including one end symbol per record and excluding start symbols. Aggregate negative log probabilities and target counts over the corpus before dividing. Do not average sentence perplexities. At zero noise, every seed must reproduce the same clean score.

9. Required results

Measure	Report and interpret
Loss and perplexity	Preliminary validation values and curves by noise type, model and tokenizer; separate tokenizer panels.
Within-pipeline change	Noisy loss minus clean loss; report either sign. Token counts and prediction targets may change.
Context benefit	Unigram loss minus bigram or interpolated loss on the same tokenized condition. Positive means an advantage over q_1.
Coverage and fragmentation	Whitespace <UNK> rate; WordPiece [UNK] rate, pieces per original word span, continuation-piece rate and token-count change.
Corruption severity	Requested p and distributions of normalized MED, paired by original example ID.

Exclude boundary symbols from diagnostics. Divide piece counts by the number of original nonempty whitespace spans; deleted spans contribute zero pieces. Define all rates. For a zero-token output, save zero counts and an undefined rate flag rather than dividing by zero or dropping the pair.

COMPARISON LIMIT: Do not rank whitespace and WordPiece by raw perplexity. Their units, vocabularies, normalization, and context spans differ. Ratios and loss changes do not fully remove these differences. A falling score can reflect unknown-token collapse or changed targets, rather than improved robustness.

These scores measure the surprise of corrupted sequences, not semantic recovery.

RESEARCH PROJECT • ASSIGNMENT 2

Execution and verification

Build a small verified example before running the full experiment.

10. Work sequence and team responsibilities

Checkpoint	Required evidence
Student A — extend pipeline	Reuse the corpus/noise code; save deterministic paired manifests and tokenization diagnostics.
Student B — build engine	Reuse tokenizer settings; implement sparse counts, $q_1/q_2/P_mix$, validation selection, and loss aggregation.
Both students — verify	Review each other's tests; compare hand calculations with program output; rerun the 20 diagnostic pairs.
Both students — evaluate	Freeze configuration, run the grid, inspect errors, and generate figures from saved machine-readable results.

Run the frozen validation grid: 3 operations × 4 nonzero rates × 3 seeds = 36 noisy datasets, plus one clean dataset. Reuse these for both tokenizers and all three model configurations: 6 × 37 = 222 individual perplexity scores. Train on clean training data only; do not refit per seed. Report 6 clean scores and 72 noisy summary rows, each with the mean, minimum and maximum across seeds. No test grid is required here.

11. Acceptance tests

- All Assignment 1 MED and reproducibility tests still pass; p = 0 leaves the input unchanged.
- For observed and unseen histories, each smoothed distribution sums to 1 within numerical tolerance; every allowed target has positive probability.
- A hand-worked vocabulary {a, b, end} with C(h,a)=2, C(h,b)=1, C(h,end)=0 gives add-one probabilities 3/6, 2/6, and 1/6.
- Interpolation weights are nonnegative and sum to 1; mixed distributions are normalized.
- Four equally likely targets give loss 2 bits and perplexity 4; corpus aggregation matches a hand calculation.
- Unseen whitespace tokens map to <UNK>; record boundaries reset context; evaluation never changes counts or vocabulary.

12. Uncertainty and error analysis

Report the mean and minimum–maximum range across three corruption seeds, not a confidence interval. Save source-document IDs, record IDs, condition IDs, per-record negative-log-probability sums and scored-target counts for later paired document bootstrap intervals. Discuss three traceable examples, including one that challenges a simple robustness explanation.

RESEARCH PROJECT • ASSIGNMENT 2

Submission and assessment

Your report is version 0.2 of the same paper you will complete this semester.

13. Submission package

Artifact	Required content
Cumulative ACL-style PDF	4 pages of main text; references and a compact appendix may follow. Use the official ACL template.
Editable manuscript	LaTeX/Overleaf source and bibliography; condense and revise Assignment 1 into the same paper.
Repository or Colab	Extended scripts, tests, environment, configuration, README, and individual contribution record.
Results and manifests	Corpus/split hashes, paired IDs, corruption logs, per-record metrics, aggregate CSV/JSON, and figure-generation code.

Include a provisional Abstract (≤200 words), Introduction, Related Work, Data and Methods, Experimental Setup, Results and Discussion, Conclusion, Limitations, and an ethics paragraph. Condense the Assignment 1 memo into this four-page ACL draft; do not append it wholesale. Argumentative Zoning may help identify aims, background, methods and evidence; it is not a required section scheme.

14. Claims, theory, and responsible research

Read the separate Core Research Axioms sheet and cite Kaplan *et al*. when discussing scaling. Carry forward licensing, privacy, bias, compute-use, and misuse considerations. Report null or mixed results as carefully as positive findings.

15. Grading rubric

Category	Weight	Evidence
Technical implementation	40%	Correct probabilities, boundaries, tokenizer reuse, and passing tests
Experimental rigor	20%	Frozen splits/configuration, paired conditions, three seeds and complete logs
Analysis and visualization	20%	Correct units, coverage diagnostics, qualified claims and error analysis
Cumulative paper	20%	ACL format, integration of Assignment 1, citations, limitations and clear writing

RESEARCH PROJECT • ASSIGNMENT 2

Your paper after Assignment 2

You will have reusable research infrastructure and an initial classical study on validation data.

16. What you can mark complete

Mark these components complete for the preliminary classical study when tests pass and results reproduce. They remain editable. Final held-out results and confidence intervals are still pending; completion is based on evidence, not pages written.

Paper component	Status after Assignment 2	Evidence you will already have
Data and noise protocol	Complete for the baseline study	Corpus card, splits, IDs, corruption rules, seeds and realized MED
Classical methods	Complete	Tokenizer specifications, vocabulary, count models, smoothing and interpolation
Baseline experimental setup	Complete	Frozen configuration, metrics, selection policy, tests and reproducible commands
Classical results	Preliminary validation study	Clean table, noise curves, diagnostics and examples; held-out confirmation remains.
Introduction	Working draft	Specific question, motivation and provisional contribution statement
Related work	Initial foundation	N-grams, BERT/WordPiece, character noise, and scaling-law framing
Discussion and limitations	Initial evidence-based draft	Observed patterns, alternative explanations, scope and ethical-use statement
Abstract and conclusion	Provisional	Accurate account of the baseline; updated after neural experiments

17. A concrete progress check

- Your partner or instructor can reproduce one headline table or figure from your README.
- Every plotted point traces to saved data, a configuration, and paired example IDs.
- Your paper explains both what the results show and what the metrics cannot establish.
- The next experiment can reuse your corpus, noise manifests, diagnostic code, and paper source.

WHAT YOU HAVE BUILT: A reproducible classical reference study on validation data. Neural experiments reuse its pipeline and diagnostics; their scores answer different questions and are not numerically interchangeable.

RESEARCH PROJECT • ASSIGNMENT 2

Where the paper goes next

The remaining work tests whether the baseline observations matter in a neural system.

18. Remaining research milestones

Next milestone	New evidence	Paper sections strengthened
Neural representations	Paired clean/noisy BERT representations with documented alignment and pooling	Methods, Results, Discussion
Task or masked-LM evaluation	Shared downstream labels or a documented masked-token objective; appropriate neural metrics	Experimental Setup, Results, Contributions
Controlled extension	One targeted ablation, extra corpus, or adequately supported scale study; defense testing if selected	Related Work, Results, Generalization
Final analysis and release	Frozen-plan test runs, paired confidence intervals, consolidated errors, figures and audit trail	Discussion, Limitations, Ethics
Publication draft	Integrated story and final abstract/conclusion; consistent citations and ACL formatting	Entire manuscript

Complete one controlled extension in Phase 4; scaling is one option, not a required finding. If you choose a defense, measure both noisy-input benefit and clean-input harm or false positives where applicable. After the semester plan is frozen, run held-out classical and neural evaluations using objective-appropriate metrics. Conference submission and acceptance are separate from the course requirement.

19. Sources and formatting authority

Jurafsky, D., and Martin, J. H. Speech and Language Processing, 3rd ed. draft (August 19, 2026), Chapter 3: N-gram Language Models. https://web.stanford.edu/~jurafsky/slp3/

Devlin et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. https://aclanthology.org/N19-1423/

Kaplan et al. (2020). Scaling Laws for Neural Language Models. https://arxiv.org/abs/2001.08361

Hugging Face. WordPiece tokenization; Perplexity of fixed-length models. https://huggingface.co/docs/course/en/chapter6/6 and https://huggingface.co/docs/transformers/perplexity

Salesforce WikiText dataset card. https://huggingface.co/datasets/Salesforce/wikitext

Official ACL style files and template. https://github.com/acl-org/acl-style-files

Notation reference

Use this page when reading the probability and loss equations.

20. Probability and evaluation terms

Symbol	Meaning in this assignment
w and w_i	w is a candidate next token. w_i is the observed target at scored position i, including an end symbol when applicable.
h and h_i	The preceding context used for prediction; h_i is that context at position i. Context resets at each record.
k	N-gram order: 1 = unigram, 2 = bigram, 3 = trigram. The history has k−1 tokens. This is not a smoothing parameter.
C(h,w)	Number of clean-training prediction events where target w follows history h.
C(h)	Sum of C(h,v) over every allowed target v in V. For an empty history, the total training target count.
V and |V|	V is the frozen set of allowed output tokens; |V| is its size. Include unknown and end symbols, but not start symbols.
P_MLE	Maximum-likelihood estimate from relative training counts, without smoothing. Unseen targets can have probability zero.
q and q_k	q denotes a probability estimate; q_k is the add-one-smoothed order-k distribution. q_1, q_2 and q_3 use zero, one and two preceding tokens.
P_mix and λ_k	P_mix is the weighted sum of q_1, q_2 and q_3. λ_k is the weight of order k; each is nonnegative and all three sum to 1.
$P(w_i \mid h_i)$	Probability assigned to the observed target by the model being scored: q_1, q_2 or P_mix. The bar means given the context.
T and i	T is the total scored target count over retained records, including end symbols. i indexes those targets; the loss sum runs from 1 to T.
H_bits and H_nats	Average negative log probability per scored token: base-2 logs give bits/token; natural logs give nats/token.
PP	Perplexity, equal to 2 raised to H_bits. It is dimensionless but depends on the tokenization and evaluation protocol.
p and MED	p is the corruption probability per eligible original character. MED is the minimum number of unit-cost character edits; it is not p or the logged edit count.

RESEARCH PROJECT • ASSIGNMENT 2

Sampling and reproducibility protocol

Fix the sample before measuring outcomes and keep the same examples in every condition.

21. A bounded sample for Assignment 2

A full-corpus run is not required. Reuse the documented Assignment 1 training slice or justify a deterministic replacement. Use one fixed validation sample for weight selection and the noise grid. Label these results as development evidence because validation also guides choices.

Decision	Required record
Selection rule	Set the clean-record eligibility rule, ordering or sampling seed, and record cap before scoring. Use source IDs, not model outcomes, to choose examples.
Length and runtime	Set a clean-record character-length cap if needed for MED runtime. Apply it before corruption and report exclusions. Do not truncate noisy variants independently.
Sample size	Record retained records, source documents, clean characters and tokens, plus checksums. Justify the scope against available compute; do not treat 20 diagnostic pairs as a sufficient aggregate study.
Grouping	Save source-article IDs using a documented corpus heading/boundary rule. Preserve groups through sampling. If unavailable, consult the instructor; do not claim document-level intervals.
Pairing	Each clean record and all its variants retain the same parent ID. Store source-document ID, record ID and condition ID in the metrics as well as the manifest.
Empty outputs	Keep records that become empty after corruption; score the end symbol. Record zero content tokens and undefined token-rate denominators explicitly.
Reserved test split	Do not inspect test outcomes or use test data for vocabulary, model or sampling choices. Define the final test sampling rule before the semester evaluation freeze.

22. Reporting and later reuse

Compute corpus loss and perplexity separately for each seed, then report the three-seed mean and minimum–maximum range for each model, tokenizer, operation and nonzero rate. These are summaries of existing scores, not additional runs. Plot separate tokenizer panels. Preserve the classical configuration for the final held-out run.

Submit a four-page paper plus references and compact appendix, not a report matching these instructions in length.